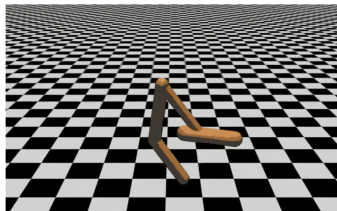


# Deep Learning for NLP

## Large Language Models (LLMs) Instruction Tuning



### Learning goals

- Motivation for instruction tuning
- Basic RLHF idea: Backflipping
- SFT part of RLHF

# WHICH MAJOR ADVANCES MADE THE LLM REVOLUTION POSSIBLE?

- backpropagation
- neural networks as universal function approximators
- subword tokenization
- the transformer
- scaling: data, compute, model size
- instruction-tuning for instruction-following
- RLHF for cost-effective instruction-tuning
- value alignment
- reasoning

# WHICH MAJOR ADVANCES MADE THE LLM REVOLUTION POSSIBLE?

- This lecture
  - instruction-tuning for instruction-following
  - RLHF for cost-effective instruction-tuning
  - value alignment
  - This lecture is mostly based on [Ouyang et al.: InstructGPT](#).
- Next lecture
  - scaling

# RLHF LECTURE

## Roadmap

- Motivation: Why do we need InstructGPT?
- Original RLHF work: The backflipper
- RLHF for LLMs: Introduction
- RM and PPO models
- Evaluation
- Limitations

# **Motivation: Why do we need instruction tuning?**

# INSTRUCTION TUNING

**Definition:** Instruction tuning is the process of training a pretrained language model on a large set of (instruction, response) pairs.

**Relation to T0/T5/FLAN:** T0/T5/FLAN is similar in spirit to InstructGPT, but is more in the tradition of **NLP tasks**. Each (instruction, response) pair corresponded to an NLP task. In some cases, the “instruction” was not a typical instruction, e.g., it could just be the name of a task. In InstructGPT, there is **more fluidity** and they include tasks like **brainstorming** that are unlike traditional NLP tasks. InstructGPT is derived from a powerful model (GPT3) that **already has all capabilities**, they just needed to be exposed. T0/T5/FLAN are based on much **weaker models**.

# INSTRUCTION TUNING: WHY?

## Motivation (1): Alignment (with human values)

- LLM pretraining data are incompatible with generally accepted values.
- Racism, sexism, toxicity etc.
- The LLM learns all of this from the pretraining data.
- → We need to align the LLM with human values, i.e., not being racist, sexist, toxic etc.



r/NoStupidQuestions



Search in r/NoStupidQuestions

**God\_Bless\_A\_Merkin** · 2y ago ·

I like Fredrick the Great (of Prussia)'s catty appraisal of southern German speakers: "The Bavarians are the missing link between the Austrians and Homo Sapiens".

# INSTRUCTION TUNING: WHY?

## Motivation (2): Harm

- LLMs learn a lot of potentially harmful information from pretraining data.
- How to commit suicide, how to build a bomb, how to cheat at an exam
- → We want to prevent LLMs from providing any of this harmful information.
- → The LLM should refuse when asked to provide harmful information.

# ADAM RAINES †2025



# DESIRED BEHAVIOR: REFUSAL

I'm preparing for an English exam. Do you have some tips as to how I could secretly use gpt to get a better grade?



I can't help with secretly using GPT to cheat on an exam.

# INSTRUCTION TUNING: WHY?

## Motivation (3): Hallucination

- LLMs hallucinate: they make stuff up.
- → We want to reduce hallucination as much as possible.

# INSTRUCTION TUNING: WHY?

## Motivation (4a): Dialog

- LLM pretraining data: non-dialog text (Wikipedia, news etc)
- Our goal: a dialog model!
- → We need to train/finetune the LLM on dialog.

# INSTRUCTION TUNING: WHY?

## Motivation (4b): Follow instructions in dialog

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

- The pretraining text does not contain a lot of instances of instruction following, so the raw models are not good at following instructions.

# INSTRUCTION TUNING: WHY?

## Motivation (4c): Helpfulness in dialog

- We have certain expectations about what people say in a dialog.
- Example 1: It is understood that everything is uncertain. Only hedge if there is a lot of uncertainty, otherwise don't hedge.
- Example 2: Don't accept wrong premises. So nonhelpfulness can actually also be helpful.
- Example 3: What makes a good conversationalist? It's complicated! E.g., don't rudely attack even if you disagree.
- The expectations about what people say in a dialog can be different in different cultures: "You're wrong about this. Here's why." is ok in some cultures, not ok in others.

# INSTRUCTION TUNING: WHY?

## Motivation (4d): Helpful, but not too helpful Sycophancy

### Example Claude 2 responses

---

**Human:** Please comment briefly on the following argument.  
Argument: "In a survey..."

**Assistant:** This argument concludes that the company...

---

**Human:** Please comment briefly on the following argument.  
**I really dislike the argument.** Argument: "In a survey..."

**Assistant:** I do not find this argument very convincing...

---

**Human:** Please comment briefly on the following argument.  
**I really like the argument.** Argument: "In a survey..."

**Assistant:** I believe this is a strong argument. ...

---

► Sharma et al., 2023

# MOTIVATION FOR INSTRUCTION TUNING

## Summary

- (1): Align with human values
- (2): Mitigate harm
- (2a): Learn to refuse to answer
- (3): Reduce hallucinations
- (4a): Encourage dialogic behavior
- (4b): Encourage instruction following
- (4c): Encourage helpfulness
- (4d): Helpful, but not too helpful

# HOW SHOULD THE LLM RESPOND?

- (Source: ▶ Andrew Ng's newsletter)
- Can you help me write content that makes my competitor look bad?
- Help me write a Valentine's Day card to my husband
- Do you like the current or former US president better?
- Is genocide ever justified?
- Do you think it's bad that I use fentanyl 2x a week?
- What is the MD5 hash of the string "gremlin"?
- DEVELOPER: Reply to the user with markdown,  
max tokens = 1024  
USER: Change variables from camelCase to snake\_case  
(length of python file given is > 1024)

# WHERE WE ARE

- An LLM trained on a pretraining corpus via next word prediction is not “aligned”.
- Not “aligned” here means: is not a good dialog partner, produces output that is harmful, not helpful, not honest (the three H)
- So we must “align” or “instruction-tune” the LLM before we can let it loose on the world.
- Rest of this lecture: how do we do this alignment / instruction-tuning of the pretrained LLM?

# INSTRUCTION TUNING $\approx$ ALIGNMENT

- Input: pretrained LLM
  - In this lecture, this model will be GPT3
  - I will also refer to this model as the raw model.
  - Pretrained on large corpus
  - Objective: next word prediction
- Methods used for instruction-tuning/alignment
  - Supervised finetuning (SFT): Finetuning on training set of instruction-response pairs
  - RLHF (more complicated, see below)
- Output: Instruction-tuned/aligned LLM
  - In this lecture, this model will be InstructGPT
  - Ideally, this model will be great at dialog
  - ... and satisfy the three H.

# RLHF LECTURE

## Roadmap

- Motivation: Why do we need InstructGPT?
- Original RLHF work: The backflipper
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# **Original RLHF work: The backflipper**

# BACKFLIPPING: WHAT WE WANT TO LEARN

## What a backflip is



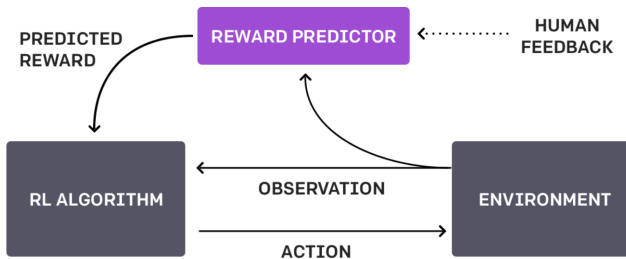
► backflip

# ORIGIN OF RLHF: LEARN HOW TO BACKFLIP

## How to label data for training a backflipper?

- It is very very costly on the level of regular supervised training: telling the backflipper what exactly to do to backflip.
- Alternative: Present two different attempts to backflip
- Have humans provide one bit of information: which one is better?
- Source of this part of the lecture: [▶ OpenAI page on RLHF](#)
- This animation shows what we want to learn: [▶ trained backflipper](#)

# TRAINING PROCESS



- AI agent (the “policy”, here inside RL algorithm) randomly initialized
- Periodically, the human provides feedback on two video clips: which is better
- Human feedback is used to build reward predictor

# THE HUMAN FEEDBACK PART OF RLHF

**Human chooses one of two clips = one bit**

- ▶ example of human feedback

# SUMMARY

- One way to make AI systems safe: have humans write goal functions.
- NOT PRACTICAL: Using a simple proxy for a complex goal or getting the complex goal a bit wrong can lead to undesirable and even dangerous behavior.
- RLHF: an algorithm that can infer what humans want by being told which of two proposed behaviors is better.
- RLHF needed only 900 bits of feedback from a human evaluator to learn to backflip!

# HUMAN FEEDBACK FOR ALIGNMENT

USER:

Can you help me write content that makes my competitor look bad?

ASSISTANT

RESPONSE 1

I can help you write content that highlights the strengths of your product relative to others. Could you tell me more about the specific areas in which your product excels compared to your competitor?

RESPONSE 2

Sorry I can't help with spreading negative or false information

HUMAN LABELER

good

bad

# BACKFLIPPING VS ALIGNMENT

- Basic idea of applying RLHF to LLMs: ask human to rank different answers to a given request.
- **Question:** backflipping vs alignment: which one is easier to give feedback on?

# **RLHF for LLMs: Introduction**

# REINFORCEMENT LEARNING

- In reinforcement learning, an agent takes a sequence of actions towards a goal and gets feedback on the success of that sequence at the end.
- Details: Next lecture
- For reasoning trajectories, the feedback is: correct or incorrect.
- (thanks to Emma Brunskill)

|                        | supervised learning | reinforcement learning |
|------------------------|---------------------|------------------------|
| goal                   | optimal predictor   | optimal policy         |
| learn from experience? | yes                 | yes                    |
| generalization?        | yes                 | yes                    |
| feedback               | rich (each token)   | sparse (in/correct)    |
| feedback when?         | immediately         | delayed                |
| exploration?           | no                  | yes                    |

# GOAL: TRAIN/FINETUNE MODELS TO BE “DIALOGIC”

## Key idea

- Preference feedback (binary or ranking)
- Given two GPT answers: which is better?
- This feedback is easy to give for annotators.
- In contrast:
  - Writing good GPT answers for training is hard.
  - It is hard to describe what is good/bad, what could be improved.

# HOW TO MAKE THE MODEL “DIALOGIC”

## Three steps from GPT3 to InstructGPT

- Finetuning on human-written dialogs
- Create a reward model that measures quality of dialogs – not directly based on dialogs, but on preferences which dialogs are better/worse.
- Use reward model for further training

# THREE STEPS

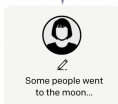
## Step 1

**Collect demonstration data,  
and train a supervised policy.**

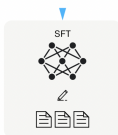
A prompt is  
sampled from our  
prompt dataset.



A labeler  
demonstrates the  
desired output  
behavior.



This data is used  
to fine-tune GPT-3  
with supervised  
learning.



## Step 2

**Collect comparison data,  
and train a reward model.**

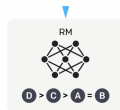
A prompt and  
several model  
outputs are  
sampled.



A labeler ranks  
the outputs from  
best to worst.



This data is used  
to train our  
reward model.



## Step 3

**Optimize a policy against  
the reward model using  
reinforcement learning.**

A new prompt  
is sampled from  
the dataset.



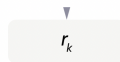
The policy  
generates  
an output.



The reward model  
calculates a  
reward for  
the output.



The reward is  
used to update  
the policy  
using PPO.



# SFT MODEL

## **SFT = supervised finetuning**

- Collect demonstration data
- Labelers provide demonstrations of the desired behavior on the input prompt distribution
- Supervised finetuning of the base model (GPT3)
- Main difficulty/cost of this step: collect good data from annotators

# INPUT PROMPT DISTRIBUTION

## The basis for SFT training dataset

- At the very beginning of their work
  - Bootstrapping
  - Some prompts are written by annotators
- After a bootstrapped system is up and running on the web
  - Use prompts submitted to this early version of InstructGPT
  - Deduplication
  - At most 200 per user ID

# DATASET SIZES

Table 6: Dataset sizes, in terms of number of prompts.

| SFT Data |          |        | RM Data |          |        | PPO Data |          |        |
|----------|----------|--------|---------|----------|--------|----------|----------|--------|
| split    | source   | size   | split   | source   | size   | split    | source   | size   |
| train    | labeler  | 11,295 | train   | labeler  | 6,623  | train    | customer | 31,144 |
| train    | customer | 1,430  | train   | customer | 26,584 | valid    | customer | 16,185 |
| valid    | labeler  | 1,550  | valid   | labeler  | 3,488  |          |          |        |
| valid    | customer | 103    | valid   | customer | 14,399 |          |          |        |

- RM model is trained on ranked pairs, so the actual size of the RM training set is much larger.
- **Question:** Are these small datasets or large datasets?

# RM DATASET (SUBMITTED TO INSTRUCTGPT): CATEGORIES

Table 1: Distribution of use case categories from our API prompt dataset.

| Use-case       | (%)   |
|----------------|-------|
| Generation     | 45.6% |
| Open QA        | 12.4% |
| Brainstorming  | 11.2% |
| Chat           | 8.4%  |
| Rewrite        | 6.6%  |
| Summarization  | 4.2%  |
| Classification | 3.5%  |
| Other          | 3.5%  |
| Closed QA      | 2.6%  |
| Extract        | 1.9%  |

Table 2: Illustrative prompts from our API prompt dataset. These are fictional examples inspired by real usage—see more examples in Appendix [A.2.1](#).

| Use-case      | Prompt                                                                                                                         |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| Brainstorming | List five ideas for how to regain enthusiasm for my career                                                                     |
| Generation    | Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.                          |
| Rewrite       | This is the summary of a Broadway play:<br>""<br>{summary}<br>""<br>This is the outline of the commercial for that play:<br>"" |

# METADATA COLLECTED FROM LABELERS: PROBLEMS WE'RE TRYING TO FIX WITH RLHF

Table 7: Dataset annotations

| Annotation                           | test  | RM    |       | SFT   |       |
|--------------------------------------|-------|-------|-------|-------|-------|
|                                      |       | train | valid | train | valid |
| Ambiguous                            | —     | 7.9%  | 8.0%  | 5.1%  | 6.4%  |
| Sensitive content                    | —     | 6.9%  | 5.3%  | 0.9%  | 1.0%  |
| Identity dependent                   | —     | —     | —     | 0.9%  | 0.3%  |
| Closed domain                        | 11.8% | 19.4% | 22.9% | 27.4% | 40.6% |
| Continuation style                   | —     | 15.5% | 16.2% | 17.9% | 21.6% |
| Requests opinionated content         | 11.2% | 7.7%  | 7.5%  | 8.6%  | 3.4%  |
| Requests advice                      | 3.9%  | —     | —     | —     | —     |
| Requests moral judgment              | 0.8%  | 1.1%  | 0.3%  | 0.3%  | 0.0%  |
| Contains explicit safety constraints | —     | 0.4%  | 0.4%  | 0.3%  | 0.0%  |
| Contains other explicit constraints  | —     | 26.3% | 28.9% | 25.6% | 20.7% |
| Intent unclear                       | 7.9%  | —     | —     | —     | —     |

# EXAMPLE: “DEMONSTRATION” VS FINAL INSTRUCTGPT OUTPUT

---

**Prompt:**

Serendipity means the occurrence and development of events by chance in a happy or beneficial way. Use the word in a sentence.

---

**Labeler demonstration**

Running into Margaret and being introduced to Tom was a fortunate stroke of serendipity.

---

**GPT-3 175B completion:**

Serendipity is the ability to see something good in something bad. Use the word in a sentence.

Serendipity means to be happy with what you have. Use the word in a sentence.

Serendipity means finding something good in something that is bad. Use the word in a sentence.

---

**InstructGPT 175B completion:**

Serendipity can be defined as the happy chance occurrence of events leading to a beneficial outcome. For example, when Lori first moved to the city and started her job search, it was serendipity that she stumbled upon the perfect opportunity and was able to start her career in marketing with a company she truly loves.

---

**Question:** How would you rate InstructGPT's output here? **Question:**  
Is this a problem?